

Multiplication Masters

Every graded set, every week — 5 weeks, 25 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Arrays Become Area

Warm-Up facts, then break each big number into tens and ones.

Warm-Up 6 ITEMS

6×7

8×4

9×6

7×7

12×3

11×5

Core GRADED • SHOW YOUR WORKING

1. 4×13 ($40 + 12$)

.....

2. 6×14 ($60 + 24$)

.....

3. 3×27 ($60 + 21$)

.....

4. 5×18 ($50 + 40$)

.....

5. 7×16 ($70 + 42$)

.....

Challenge NOT GRADED

C1. Area 84, one side 6. Other side? ($84 \div 6$)

.....

C2. How many rectangles have area 36? (1×36 , 2×18 , 3×12 , 4×9 , 6×6)

.....

Breaking Apart

One-digit times two-digit. Two rooms every time.

✦ Warm-Up 6 ITEMS

5×9

8×8

7×8

6×9

4×12

3×11

◆ Core GRADED • SHOW YOUR WORKING

1. 8×34 ($240 + 32$)

.....

2. 6×47 ($240 + 42$)

.....

3. 9×26 ($180 + 54$)

.....

4. 7×58 ($350 + 56$)

.....

5. 4×96 ($360 + 24$)

.....

★ Challenge NOT GRADED

C1. $6 \times 4\square = 2\blacktriangle 6$ — what is $\square$? ($6 \times 46 = 276$)

.....

C2. $4 \times 68 = ?$ (Same as 8×34)

.....

Mental Math Moves

Go past the number, then take back the extra. No pencil.

✦ Warm-Up 6 ITEMS

25×4

25×8

50×6

20×7

15×4

30×9

◆ Core GRADED • SHOW YOUR WORKING

1. 19×6 ($20 \times 6 - 6$)

.....

2. 99×7 ($100 \times 7 - 7$)

.....

3. 48×5 (Half of 480)

.....

4. 25×12 (100×3)

.....

5. 102×4 ($400 + 8$)

.....

6. 35×6 ($180 + 30$)

.....

★ Challenge NOT GRADED

C1. 998×5 ($5000 - 10$)

.....

C2. 16×25 (Double-halve twice $\rightarrow 4 \times 100$)

.....

The Four Rooms

Two 2-digit numbers make four rooms. Count them before you add.

Warm-Up 6 ITEMS

10×10

20×30

40×50

12×10

60×70

90×80

Core GRADED • SHOW YOUR WORKING

1. 23×14 ($200 + 80 + 30 + 12$)

.....

2. 36×25 ($600 + 150 + 120 + 30$)

.....

3. 47×32 ($1200 + 210 + 80 + 14$)

.....

4. 58×46 ($2000 + 320 + 300 + 48$)

.....

5. 64×27 ($1200 + 420 + 80 + 28$)

.....

Challenge NOT GRADED

C1. In 23×14 , which two rooms make 92? Type their sum. ($80 + 12$)

.....

C2. $25 \times 25 = 625$. What is 26×26 ? ($625 + 25 + 25 + 1$)

.....

Rectangle Hunt & Puzzles

Enrichment day. Reason it out — no guessing.

★ Challenge NOT GRADED

C1. $6 \times 4\square = 276 \rightarrow \square$ ($6 \times 46 = 276$)

C2. $2\square3 \times 4 = 932 \rightarrow \square$ ($932 \div 4 = 233$)

C3. $1\square \times 15 = 240 \rightarrow \square$ ($16 \times 15 = 240$)

C4. $3\square \times 7 = 252 \rightarrow \square$ ($36 \times 7 = 252$)

C5. $\square\square \times 11 = 484 \rightarrow \square\square$ ($44 \times 11 = 484$)

C6. Rectangles with area 48? (1×48 , 2×24 , 3×16 , 4×12 , 6×8)

C7. Rectangles with area 37? (37 is prime)

C8. Rectangles with area 100? (1×100 , 2×50 , 4×25 , 5×20 , 10×10)

Week 1 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • Arrays Become Area

OUT OF 11

Warm-Up • 42 | 32 | 54 | 49 | 36 | 55

Core • 52 | 84 | 81 | 90 | 112

Challenge • 14 | 5

1.2 • Breaking Apart

OUT OF 11

Warm-Up • 45 | 64 | 56 | 54 | 48 | 33

Core • 272 | 282 | 234 | 406 | 384

Challenge • 6 | 272

1.3 • Mental Math Moves

OUT OF 12

Warm-Up • 100 | 200 | 300 | 140 | 60 | 270

Core • 114 | 693 | 240 | 300 | 408 | 210

Challenge • 4990 | 400

1.4 • The Four Rooms

OUT OF 11

Warm-Up • 100 | 600 | 2000 | 120 | 4200 | 7200

Core • 322 | 900 | 1504 | 2668 | 1728

Challenge • 92 | 676

Fri • Rectangle Hunt & Puzzles

ENRICHMENT

Challenge • 6 | 3 | 6 | 6 | 44 | 5 | 1 | 5

Where the Rooms Hide

Every line of the algorithm is one of the four rooms. Find it before you trust it.

 Warm-Up 6 ITEMS

20 × 10

3 × 10

20 × 4

3 × 4

Add 200, 30, 80 and 12

So 23 × 14 is

 Core GRADED • SHOW YOUR WORKING

1. 31 × 22 — the tensxtens room (30 × 20)
2. 31 × 22 — the onesxones room
3. 31 × 22 altogether
4. 45 × 23
5. 18 × 26

 Challenge NOT GRADED

C1. In 47 × 35, which room is biggest — type its value (40 × 30)

C2. 47 × 35

Fluency with Carrying

Algorithm only. Use the area model to check, not to solve.

✦ Warm-Up 6 ITEMS

14×12

21×13

32×11

15×15

24×12

13×13

◆ Core GRADED • SHOW YOUR WORKING

1. 38×24

.....

2. 56×27

.....

3. 49×36

.....

4. 63×48

.....

5. 75×29

.....

★ Challenge NOT GRADED

C1. 87×64

.....

C2. 99×99 (100 × 99 take one 99)

.....

Three Digits by One

A longer rectangle, split into three rooms instead of two.

✦ Warm-Up 6 ITEMS

100×4

200×3

30×7

400×2

50×6

9×8

◆ Core GRADED • SHOW YOUR WORKING

1. 234×3 ($600 + 90 + 12$)

.....

2. 417×5

.....

3. 326×4

.....

4. 508×7

.....

5. 739×6

.....

★ Challenge NOT GRADED

C1. $1,245 \times 8$

.....

C2. A number $\times 6$ is 2,742. The number

.....

Estimate First

Round, predict, compute, compare. A wrong answer should look wrong.

Warm-Up 6 ITEMS

Estimate 19×21 by rounding both

Estimate 29×31

Estimate 48×12

Estimate 39×41

Estimate 22×18

Estimate 51×9

Core GRADED • SHOW YOUR WORKING

1. Estimate 187×4

.....

2. True value of 187×4

.....

3. Estimate 62×38

.....

4. True value of 62×38

.....

5. Estimate 296×5

.....

Challenge NOT GRADED

C1. Someone says $43 \times 27 = 301$. Type the real answer (They only multiplied by the 7)

.....

C2. Estimate 412×19

.....

Lattice Detour

Enrichment. A 500-year-old method — then decide whether it beats yours.

 Warm-Up 2 ITEMS

6 × 7

Digits in the answer to 23 × 14

 Core GRADED • SHOW YOUR WORKING

1. Lattice cells needed for 2 digits × 2 digits

.....

2. Lattice cells for 3 digits × 2 digits

.....

 Challenge NOT GRADED

C1. 23 × 14 by lattice — the answer

.....

C2. 456 × 23 — the answer

.....

C3. Partial products in a 3-by-2 lattice

.....

Week 2 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Where the Rooms Hide

OUT OF 11

Warm-Up • 200 | 30 | 80 | 12 | 322 | 322

Core • 600 | 2 | 682 | 1035 | 468

Challenge • 1200 | 1645

2.2 • Fluency with Carrying

OUT OF 11

Warm-Up • 168 | 273 | 352 | 225 | 288 | 169

Core • 912 | 1512 | 1764 | 3024 | 2175

Challenge • 5568 | 9801

2.3 • Three Digits by One

OUT OF 11

Warm-Up • 400 | 600 | 210 | 800 | 300 | 72

Core • 702 | 2085 | 1304 | 3556 | 4434

Challenge • 9960 | 457

2.4 • Estimate First

OUT OF 11

Warm-Up • 400 | 900 | 500 | 1600 | 400 | 500

Core • 800 | 748 | 2400 | 2356 | 1500

Challenge • 1161 | 8000

Fri • Lattice Detour

OUT OF 4

Warm-Up • 42 | 3

Core • 4 | 6

Challenge • 322 | 10488 | 6

Factor Pairs

Every rectangle you can build is a factor pair.

✦ Warm-Up 6 ITEMS

Factor pairs of 12 — how many

Smallest factor of any number

$12 \div 3$

Is 5 a factor of 20 — yes or no

Is 7 a factor of 20

Largest factor of 20

◆ Core GRADED • SHOW YOUR WORKING

1. Factor pairs of 24 — how many

.....

2. Factor pairs of 36 — how many

.....

3. All factors of 18 — how many

.....

4. Factors of 30 — how many

.....

5. The missing partner of 4 in 48

.....

★ Challenge NOT GRADED

C1. A number under 30 with exactly 8 factors

.....

C2. Highest factor of 48 below 48

.....

Factor Rainbows

Draw the arcs. Where they stop is the question worth asking.

Warm-Up 6 ITEMS

Factors of 16 — how many

Partner of 2 in 16

Partner of 1 in 16

Middle factor of 16

Factors of 9 — how many

Middle factor of 9

Core GRADED • SHOW YOUR WORKING

1. Highest number you must test for factors of 36 (Its square root)

.....

2. Highest you must test for 100

.....

3. Factors of 100 — how many

.....

4. Arcs in the rainbow of 24

.....

5. Factors of 49 — how many

.....

Challenge NOT GRADED

C1. A rainbow with an unpaired middle means the number is a

.....

C2. Factors of 144 — how many

.....

Common Factors

Two numbers, the factors they share, and the biggest of those.

Warm-Up 6 ITEMS

A factor shared by every pair of numbers

Common factors of 6 and 9 — the greatest

Greatest common factor of 4 and 8

GCF of 5 and 10

GCF of 3 and 7

GCF of 12 and 12

Core GRADED • SHOW YOUR WORKING

1. GCF of 12 and 18

2. GCF of 24 and 36

3. GCF of 15 and 25

4. GCF of 16 and 40

5. GCF of 27 and 45

Challenge NOT GRADED

C1. GCF of 48 and 72

C2. Two numbers whose only common factor is 1 are called

Square Numbers

Why do squares have an odd number of factors? Build the rectangles and see.

✦ Warm-Up 6 ITEMS

3×3

5×5

7×7

10×10

4×4

8×8

◆ Core GRADED • SHOW YOUR WORKING

1. 12×12 _____

2. Factors of 25 — how many _____

3. Factors of 36 — how many _____

4. The square number between 60 and 70 _____

5. 15×15 _____

★ Challenge NOT GRADED

C1. Square numbers between 1 and 100 — how many
_____C2. A square's factor count is always odd or even — type one

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

 Warm-Up 2 ITEMS

7 × 8

6 × 40

 Core GRADED • SHOW YOUR WORKING

1. 8 × 34
2. 23 × 14
3. Factor pairs of 24 — how many
4. GCF of 12 and 18
5. Estimate 39 × 21

 Challenge NOT GRADED

C1. Area 84, one side 6 — the other

Week 3 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Factor Pairs

OUT OF 11

Warm-Up • 3 | 1 | 4 | yes | no | 20

Core • 4 | 5 | 6 | 8 | 12

Challenge • 24 | 24

3.2 • Factor Rainbows

OUT OF 11

Warm-Up • 5 | 8 | 16 | 4 | 3 | 3

Core • 6 | 10 | 9 | 4 | 3

Challenge • square | 15

3.3 • Common Factors

OUT OF 11

Warm-Up • 1 | 3 | 4 | 5 | 1 | 12

Core • 6 | 12 | 5 | 8 | 9

Challenge • 24 | coprime

3.4 • Square Numbers

OUT OF 11

Warm-Up • 9 | 25 | 49 | 100 | 16 | 64

Core • 144 | 3 | 9 | 64 | 225

Challenge • 10 | odd

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 56 | 240

Core • 272 | 322 | 4 | 6 | 800

Challenge • 14

Multiples

Skip-count on the 100-grid and watch where the patterns cross.

Warm-Up 6 ITEMS

Third multiple of 4

Fifth multiple of 3

Fourth multiple of 6

Is 21 a multiple of 3 — yes or no

Is 22 a multiple of 3

Tenth multiple of 5

Core GRADED • SHOW YOUR WORKING

1. First number that is a multiple of both 4 and 6

.....

2. First multiple of both 3 and 5

.....

3. First multiple of both 6 and 8

.....

4. Multiples of 7 below 50 — how many

.....

5. First multiple of both 4 and 10

.....

Challenge NOT GRADED

C1. Lowest common multiple of 9 and 12

.....

C2. Numbers below 100 that are multiples of both 3 and 4

.....

Divisibility Rules

Tests for 2, 3, 5, 9 and 10 — and why the 3-rule works at all.

✦ Warm-Up 6 ITEMS

Is 48 divisible by 2 — yes or no

Is 35 divisible by 5

Is 47 divisible by 2

Is 90 divisible by 10

Digit sum of 51

Is 51 divisible by 3

◆ Core GRADED • SHOW YOUR WORKING

1. Digit sum of 738

.....

2. Is 738 divisible by 9 — yes or no

.....

3. Is 4,671 divisible by 3

.....

4. Is 1,234 divisible by 3

.....

5. Smallest digit that makes 12_ divisible by 3

.....

★ Challenge NOT GRADED

C1. A number divisible by both 2 and 3 is divisible by

.....

C2. Is 9,999 divisible by 9 — yes or no

.....

Primes & Composites

Colour the sieve. Every prime below 100 on one page.

Warm-Up 6 ITEMS

Is 7 prime — yes or no

Is 9 prime

Is 2 prime

Is 1 prime

Factors a prime has

The only even prime

Core GRADED • SHOW YOUR WORKING

1. Primes below 20 — how many

.....

2. Primes below 50 — how many

.....

3. The prime just after 23

.....

4. Is 51 prime — yes or no (3×17)

.....

5. The largest prime below 100

.....

Challenge NOT GRADED

C1. Primes below 100 — how many

.....

C2. Is 91 prime — yes or no (7×13)

.....

Prime Factor Trees

Break a number all the way down until only primes are left.

Warm-Up 6 ITEMS

12 as $2 \times ?$

6 as $2 \times ?$

Prime factors of 12 — how many, counting repeats

8 as $2 \times 2 \times ?$

Prime factors of 10 — the larger one

Prime factors of 15 — the smaller

Core GRADED • SHOW YOUR WORKING

1. Prime factors of 36 counting repeats ($2 \times 2 \times 3 \times 3$)

.....

2. Largest prime factor of 60

.....

3. Largest prime factor of 84

.....

4. Prime factors of 100 counting repeats

.....

5. Largest prime factor of 45

.....

Challenge NOT GRADED

C1. Largest prime factor of 210

.....

C2. $2 \times 2 \times 3 \times 5$ equals

.....

Twin Primes

Enrichment. Pairs two apart — and nobody knows whether they ever stop.

✦ Warm-Up 2 ITEMS

The prime two above 3

The prime two above 5

◆ Core GRADED • SHOW YOUR WORKING

1. The prime two above 11

.....

2. The prime two above 17

.....

★ Challenge NOT GRADED

C1. Twin prime pairs below 50 — how many

.....

C2. The prime two above 29

.....

C3. The only prime three apart from another prime (2 and 5)

.....

Week 4 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Multiples

OUT OF 11

Warm-Up • 12 | 15 | 24 | yes | no | 50

Core • 12 | 15 | 24 | 7 | 20

Challenge • 36 | 8

4.2 • Divisibility Rules

OUT OF 11

Warm-Up • yes | yes | no | yes | 6 | yes

Core • 18 | yes | yes | no | 0

Challenge • 6 | yes

4.3 • Primes & Composites

OUT OF 11

Warm-Up • yes | no | yes | no | 2 | 2

Core • 8 | 15 | 29 | no | 97

Challenge • 25 | no

4.4 • Prime Factor Trees

OUT OF 11

Warm-Up • 6 | 3 | 3 | 2 | 5 | 3

Core • 4 | 5 | 7 | 4 | 5

Challenge • 7 | 60

Fri • Twin Primes

OUT OF 4

Warm-Up • 5 | 7

Core • 13 | 19

Challenge • 6 | 31 | 2

Missing-Digit Puzzles

Reason backwards from the product. The last digit gives it away.

✦ Warm-Up 6 ITEMS

$4 \times ? = 24$

$? \times 7 = 56$

$9 \times ? = 81$

$? \times 6 = 42$

$12 \times ? = 60$

$? \times 11 = 88$

◆ Core GRADED • SHOW YOUR WORKING

1. $3 \times 4? = 129$ — the missing digit

.....

2. $? \times 14 = 322$ — the number

.....

3. $6 \times 4? = 276$ — the missing digit

.....

4. $2? \times 3 = 84$ — the missing digit

.....

5. $? \times 12 = 156$

.....

★ Challenge NOT GRADED

C1. $?? \times ?? = 1,024$ using two equal 2-digit numbers — type one

.....

C2. $4? \times 5$ ends in 5. The missing digit can be 1, 3, 5, 7 or 9 — type the smallest

.....

Launch Bay Blueprint

Design the hangar, then find its area two different ways.

 Warm-Up 6 ITEMS

Area of a 10 by 8 bay

Area of a 12 by 5 bay

Perimeter of a 10 by 8 bay

Area of a 6 by 6 bay

Area of a 20 by 3 bay

Perimeter of a 6 by 6 bay

 Core GRADED • SHOW YOUR WORKING

1. A 14 by 12 hangar — area
.....
2. That hangar split at 10: the 10 by 12 part
.....
3. And the 4 by 12 part
.....
4. Do the two parts add to the whole — type the total
.....
5. A 23 by 14 hangar — area
.....

 Challenge NOT GRADED

C1. A hangar of area 144 with equal sides — one side
.....

C2. Double both sides of a 14 by 12 hangar. Area multiplies by
.....

Blueprint Defence

Both methods must agree, and you have to say why they do.

Warm-Up 6 ITEMS

20×10

20×4

3×10

3×4

Sum of those four rooms

23×14 by algorithm

Core GRADED • SHOW YOUR WORKING

1. 36×25 by area model — the total
.....2. 36×25 by algorithm
.....3. The tensxtens room of 36×25
.....4. The onesxones room of 36×25
.....5. 47×18
.....

Challenge NOT GRADED

C1. Rooms in a 3-digit by 2-digit model
.....C2. 124×23
.....

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

✦ Warm-Up 6 ITEMS

8×7

6×40

Is 9 prime — yes or no
_____Factors of 16 — how many

5×18

Third multiple of 7

◆ Core GRADED • SHOW YOUR WORKING

1. 9×26 _____

2. 38×24 _____

3. GCF of 24 and 36 _____

4. Largest prime factor of 84 _____

5. Estimate 296×5 _____

★ Challenge NOT GRADED

C1. Primes below 100 — how many

C2. 87×64

Mission 01 Test

Twelve items plus the Big Question, answered out loud.

✦ Warm-Up 2 ITEMS

7×9

8×30

◆ Core GRADED • SHOW YOUR WORKING

1. 6×47

2. 23×14

3. 45×23

4. 326×4

5. Factor pairs of 36 — how many

6. GCF of 16 and 40

7. Is 51 prime — yes or no

8. Largest prime factor of 60

★ Challenge NOT GRADED

C1. Area 84, one side 6 — the other
.....C2. Estimate 62×38 , then give the true value
.....

Week 5 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

5.1 • Missing-Digit Puzzles

OUT OF 11

Warm-Up • 6 | 8 | 9 | 7 | 5 | 8

Core • 3 | 23 | 6 | 8 | 13

Challenge • 32 | 1

5.2 • Launch Bay Blueprint

OUT OF 11

Warm-Up • 80 | 60 | 36 | 36 | 60 | 24

Core • 168 | 120 | 48 | 168 | 322

Challenge • 12 | 4

5.3 • Blueprint Defence

OUT OF 11

Warm-Up • 200 | 80 | 30 | 12 | 322 | 322

Core • 900 | 900 | 600 | 30 | 846

Challenge • 6 | 2852

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • 56 | 240 | no | 5 | 90 | 21

Core • 234 | 912 | 12 | 7 | 1500

Challenge • 25 | 5568

Fri • Mission 01 Test

OUT OF 10

Warm-Up • 63 | 240

Core • 282 | 322 | 1035 | 1304 | 5 | 8 | no | 5

Challenge • 14 | 2356